

Given: Due:	Further Maths (Core Pure) HW 07	Name: MARK SCHEME
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Chapter 6 (6.1 – 6.4): Matrix arithmetic & 2x2 inverses

1) Given that k is a real number and that

$$\mathbf{A} = \begin{pmatrix} 1+k & k \\ k & 1-k \end{pmatrix}$$

find the exact values of k for which $\mathbf{A}$ is a singular matrix. Give your answers in their simplest form.

(3)

The majority of students knew the definition of a singular matrix and got this question completely correct. A few were unable to find determinant accurately, but many found the correct determinant and equated to zero and succeeded in obtaining a fully correct solution. The most commonly seen errors were, having arrived correctly at $k^2 = \frac{1}{2}$, proceeding to give the positive root only. Unfortunately, $k^2 = \frac{1}{2}$ producing a solution of $k = \pm \frac{1}{2}$ was not a rare occurrence.

Question Number	Scheme	Marks
	Determinant of $\mathbf{A} = (1-k)(1+k) - k^2 = 0$	M1
	$1 - k + k - k^2 - k^2 (= 0)$ $1 - 2k^2 (= 0)$ So $k = \pm \frac{\sqrt{2}}{2}$	A1 A1 (3)
		3 marks
Notes M1: for attempting $ad - bc = 0$ with '$= 0$' seen or used later in the solution. A1: Correct (unsimplified) expression on LHS or correct equation after brackets expanded. A1: Accept $\pm \frac{\sqrt{2}}{2}$, $\pm \frac{1}{\sqrt{2}}$, $\pm \sqrt{\frac{1}{2}}$, $\pm \sqrt{0.5}$. Must have $\pm$ for mark.		

2)

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}$$

Given that $\mathbf{M} = (\mathbf{A} + \mathbf{B})(2\mathbf{A} - \mathbf{B})$,

(a) calculate the matrix $\mathbf{M}$,

(6)

(b) find the matrix $\mathbf{C}$ such that $\mathbf{MC} = \mathbf{A}$.

(4)

This was expected to be a good source of marks for all students, but unfortunately accuracy was an issue for some students. In question (a) sums, differences and/or products of matrices often had inaccurate elements which lost at least two marks. In question (b) a number of students did not take the importance of the order of matrix multiplication into account and a large number could not find an inverse of a 2 x2 matrix accurately. Those students who decided to use simultaneous equations were less likely to produce an accurate answer due to accuracy errors.



Question Number	Scheme	EXAM PAPERS PRACTICE	Marks
(a)	$A + B = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$	M1: Correct attempt at matrix addition with 3 elements correct A1: Correct matrix	M1A1
	$2A - B = \begin{pmatrix} 5 & 1 \\ -2 & -1 \end{pmatrix}$	M1: Correct attempt to double A and subtract B 3 elements correct A1: Correct matrix	M1A1
	$(A + B)(2A - B) = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ -2 & -1 \end{pmatrix}$		
	$\begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ -2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -7 & -2 \end{pmatrix}$	M1: Correct method to multiply A1: cao	M1A1
			(6)
(a) Way 2	$(A + B)(2A - B) = 2A^2 + 2BA - AB - B^2$	M1: Expands brackets with at least 3 correct terms A1: Correct expansion	M1A1
	$A^2 = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix}, BA = \begin{pmatrix} -3 & -1 \\ -1 & 0 \end{pmatrix},$ $AB = \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}, B^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	M1: Attempts A^2 , B^2 and AB or BA A1: Correct matrices	M1A1
	$2A^2 + 2BA - AB - B^2 = \begin{pmatrix} 1 & -1 \\ -7 & -2 \end{pmatrix}$	M1: Substitutes into their expansion A1: Correct matrix	M1A1
(b)	$MC = A \Rightarrow C = M^{-1}A$	May be implied by later work	B1
	$M^{-1} = \frac{1}{-2-7} \begin{pmatrix} -2 & 1 \\ 7 & 1 \end{pmatrix}$	An attempt at their $\frac{1}{\det M} \begin{pmatrix} -2 & 1 \\ 7 & 1 \end{pmatrix}$	M1
	$C = \frac{1}{-2-7} \begin{pmatrix} -2 & 1 \\ 7 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$	Correct order required and an attempt to multiply	dM1
	$C = -\frac{1}{9} \begin{pmatrix} -5 & -2 \\ 13 & 7 \end{pmatrix}$	oe	A1
			(4)
			Total 10
(b) Way 2	$\begin{pmatrix} 1 & -1 \\ -7 & -2 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$	Correct statement	B1
	$a - c = 2, b - d = 1$ $-7a - 2c = -1, -7b - 2d = 0$	Multiplies correctly to obtain 4 equations	M1
	$a = \frac{5}{9}, b = \frac{2}{9}, c = -\frac{13}{9}, d = -\frac{7}{9}$	M1: Solves to obtain values for a, b, c and d A1: Correct values	M1A1

3) (i) Given that

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 4 & 5 \end{pmatrix} \text{ and } B = \begin{pmatrix} 2 & -1 & 4 \\ 1 & 3 & 1 \end{pmatrix},$$

(a) find **AB**.

(b) Explain why **AB** $\neq$ **BA**.

(4)

(ii) Given that

$$C = \begin{pmatrix} 2k & -2 \\ 3 & k \end{pmatrix}, \text{ where } k \text{ is a real number}$$

find **C**⁻¹, giving your answer in terms of *k*.

(3)

In question (i)(a) there were a lot of correct answers with almost all students gaining at least one mark, although there were some who made two arithmetic mistakes which lost both accuracy marks. A few students failed to get the correct dimensional order of the answer.

In question (b) explanations for why **AB** and **BA** are not equal were many and varied and often correct. There were lots of references to transformations and matrix multiplication not being commutative, but these attempts were not usually sufficient for this mark. The most common approach was a discussion about the dimensions of the resulting matrices.

Question (c) was generally very well done, with the majority of answers gaining full marks. There were some errors in the calculation of the determinant and also some errors in the positions and signs of the elements within the inverse matrix. A significant number of students took the time to either factorise the determinant or to multiply the determinant into the matrix, neither of which were necessary to get all the available marks.

Question Number	Scheme	Notes	Marks
	$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 4 & 5 \end{pmatrix}, B = \begin{pmatrix} 2 & -1 & 4 \\ 1 & 3 & 1 \end{pmatrix}$		
(i)(a)	$\begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 2 & -1 & 4 \\ 1 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & -6 & 11 \\ 13 & 11 & 21 \end{pmatrix}$	M1: 3x3 matrix with a number or numerical expression for each element A2:cao (-1 each error) Only 1 error award A1A0	M1A2
(b)	$BA = \begin{pmatrix} 2 & -1 & 4 \\ 1 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 15 & 25 \\ 14 & 4 \end{pmatrix}$	Allow any convincing argument. E.g.s BA is a 2x2 matrix (so AB ≠ BA) or dimensionally different. Attempt to evaluate product not required. NB 'Not commutative' only is B0	B1
			(4)
(ii)	$(\det C) = 2k \times k - 3 \times (-2)$	Correct attempt at determinant	M1
	$C^{-1} = \frac{1}{2k^2 + 6} \begin{pmatrix} k & 2 \\ -3 & 2k \end{pmatrix}$	M1: $\frac{1}{\text{their det } C} \begin{pmatrix} k & 2 \\ -3 & 2k \end{pmatrix}$ A1:cao oe	M1A1
			(3)
			Total 7

4)

$$M = \begin{pmatrix} x & x-2 \\ 3x-6 & 4x-11 \end{pmatrix}$$

Given that the matrix **M** is singular, find the possible values of x.

(4)

The majority of candidates understood what was meant by a singular matrix and gained full marks for this question. The usual sign errors caused some to lose marks and some factorised incorrectly. There were a significant number of candidates who wrote down the solutions to their quadratic with no working shown, possibly using a graphic calculator. Some put the determinant equal to 1 and some went on to find the inverse matrix.

Question Number	Scheme	Notes	Marks
	$M = \begin{pmatrix} x & x-2 \\ 3x-6 & 4x-11 \end{pmatrix}$		
	$\det M = x(4x-11) - (3x-6)(x-2)$	Correct attempt at determinant	M1
	$x^2 + x - 12 (=0)$	Correct 3 term quadratic	A1
	$(x+4)(x-3) (=0) \rightarrow x = \dots$	Their 3TQ = 0 and attempts to solve relevant quadratic using factorisation or completing the square or correct quadratic formula leading to x =	M1
	$x = -4, x = 3$	Both values correct	A1
			(4)
			Total 4
Notes			
	$x(4x-11) = (3x-6)(x-2)$ award first M1 $\neq (x^2 + x - 12)$ seen award first M1A1		
	Method mark for solving 3 term quadratic: 1. <u>Factorisation</u> $(x^2 + bx + c) = (x+p)(x+q)$, where $ pq = c $, leading to x = $(ax^2 + bx + c) = (mx+p)(nx+q)$, where $ pq = c $ and $ nm = a $, leading to x = 2. <u>Formula</u> Attempt to use <u>correct</u> formula (with values for a, b and c). 3. <u>Completing the square</u> Solving $x^2 + bx + c = 0$: $\left(x \pm \frac{b}{2}\right)^2 \pm q \pm c$, $q \neq 0$, leading to x = ... Both correct with no working 4/4, only one correct 0/4		

5)

(a) Given that

$$A = \begin{pmatrix} 3 & 1 & 3 \\ 4 & 5 & 5 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 0 & -1 \end{pmatrix}$$

find **AB**.

(2)

(b) Given that

$$C = \begin{pmatrix} 3 & 2 \\ 8 & 6 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 2k \\ 4 & k \end{pmatrix}, \text{ where } k \text{ is a constant}$$

and

$$E = C + D$$

find the value of k for which **E** has no inverse.

(4)

Part (a) was well answered with most candidates multiplying the matrices together correctly. Only a few candidates multiplied the matrices the wrong way round. In part (b) many could add the matrices correctly although a few candidates multiplied. The condition for there not being an inverse for **E** was well known and most attempted the determinant and set it to zero. The resulting equation in k was usually solved correctly although there were some basic algebraic errors.

Question Number	Scheme	Notes	Marks
(a)	$A = \begin{pmatrix} 3 & 1 & 3 \\ 4 & 5 & 5 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 0 & -1 \end{pmatrix}$		
	$AB = \begin{pmatrix} 3 & 1 & 3 \\ 4 & 5 & 5 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 0 & -1 \end{pmatrix}$		
	$= \begin{pmatrix} 3+1+0 & 3+2-3 \\ 4+5+0 & 4+10-5 \end{pmatrix}$	A correct method to multiply out two matrices. Can be implied by two out of four correct (unsimplified) elements in a dimensionally correct matrix. A 2x2 matrix with a number or a calculation at each corner.	M1
	$= \begin{pmatrix} 4 & 2 \\ 9 & 9 \end{pmatrix}$	Correct answer	A1
	A correct answer with no working can score both marks		
			[2]
(b)	$C = \begin{pmatrix} 3 & 2 \\ 8 & 6 \end{pmatrix}, D = \begin{pmatrix} 5 & 2k \\ 4 & k \end{pmatrix}, \text{ where } k \text{ is a constant,}$		
	$C + D = \begin{pmatrix} 3 & 2 \\ 8 & 6 \end{pmatrix} + \begin{pmatrix} 5 & 2k \\ 4 & k \end{pmatrix} = \begin{pmatrix} 8 & 2k+2 \\ 12 & 6+k \end{pmatrix}$	An attempt to add C to D. Can be implied by two out of four correct (unsimplified) elements in a dimensionally correct matrix.	M1
	$E \text{ does not have an inverse} \Rightarrow \det E = 0.$		
	$8(6+k) - 12(2k+2)$	Applies "ad - bc" to E where E is a 2x2 matrix.	M1
	$8(6+k) - 12(2k+2) = 0$	States or applies $\det(E) = 0$ where $\det(E) = ad - bc$ or $ad + bc$ only and E is a 2x2 matrix.	M1
	Note $8(6+k) - 12(2k+2) = 0$ or $8(6+k) = 12(2k+2)$ could score both M's		
	$48 + 8k = 24k + 24$		
	$24 = 16k$		
	$k = \frac{3}{2}$		A1 oe
			[4]
			6 marks

6)

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$$

(a) Show that **A** is non-singular.

(2)

(b) Find **B** such that $BA^2 = A$.

(4)

In part (a) the majority of candidates knew to evaluate the determinant but there were a significant number of cases where candidates thought this was sufficient with no reference to the fact that it was not zero. Part (b) was tackled in a variety of ways. Those candidates who did not appreciate that $\mathbf{B} = \mathbf{A}^{-1}$ began by evaluating $\mathbf{A}^2$ and replaced $\mathbf{B}$ with a general matrix to produce simultaneous equations and quite often went on to find $\mathbf{B}$ correctly. Some also started the same way by evaluating $\mathbf{A}^2$ and then multiplied $\mathbf{A}$ by the inverse of $\mathbf{A}^2$ to find $\mathbf{B}$.

Question Number	Scheme	Notes	Marks
(a)	$\det(A) = 3 \times 0 - 2 \times 1 (= -2)$	Correct attempt at the determinant	M1
	$\det(A) \neq 0$ (so A is non singular)	$\det(A) = -2$ and some reference to zero	A1
	$\frac{1}{\det(A)}$ scores M0		(2)
(b)	$BA^2 = A \Rightarrow BA = I \Rightarrow B = A^{-1}$	Recognising that A^{-1} is required	M1
	$B = -\frac{1}{2} \begin{pmatrix} 3 & -1 \\ -2 & 0 \end{pmatrix}$	At least 3 correct terms in $\begin{pmatrix} 3 & -1 \\ -2 & 0 \end{pmatrix}$	M1
		$\frac{1}{\text{their } \det(A)} \begin{pmatrix} * & * \\ * & * \end{pmatrix}$	B1ft
		Fully correct answer	A1
	Correct answer only score 4/4		(4)
Ignore poor matrix algebra notation if the intention is clear			Total 6
(b) Way 2	$A^2 = \begin{pmatrix} 2 & 3 \\ 6 & 11 \end{pmatrix}$	Correct matrix	B1
	$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 6 & 11 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} \Rightarrow \begin{matrix} 2a+6b=0 & 2c+6d=2 \\ 3a+11b=1 & 3c+11d=3 \end{matrix}$	2 equations in a and b or 2 equations in c and d	M1
	$a = -\frac{3}{2}, b = \frac{1}{2}, c = 1, d = 0$	M1 Solves for a and b or c and d	M1A1
		A1 All 4 values correct	
(b) Way 3	$A^2 = \begin{pmatrix} 2 & 3 \\ 6 & 11 \end{pmatrix}$	Correct matrix	B1
	$(A^2)^{-1} = \frac{1}{2 \times 11 - 3 \times 6} \begin{pmatrix} 11 & -3 \\ -6 & 2 \end{pmatrix}$ see note	Attempt inverse of A^2	M1
	$A(A^2)^{-1} = \frac{1}{4} \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 11 & -3 \\ -6 & 2 \end{pmatrix}$ or $\frac{1}{4} \begin{pmatrix} 11 & -3 \\ -6 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$	Attempts $A(A^2)^{-1}$ or $(A^2)^{-1}A$	M1
	$B = -\frac{1}{2} \begin{pmatrix} 3 & -1 \\ -2 & 0 \end{pmatrix}$	Fully correct answer	A1
(b) Way 4	$BA = I$	Recognising that $BA = I$	B1
	$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{matrix} 2b=1 & 2d=0 \\ a+3b=0 & c+3d=1 \end{matrix}$	2 equations in a and b or 2 equations in c and d	M1
	$a = -\frac{3}{2}, b = \frac{1}{2}, c = 1, d = 0$	M1 Solves for a and b or c and d	M1A1
		A1 All 4 values correct	

Extra Notes

(b) Way 3

Attempting inverse of $\mathbf{A}^2$ needs to be recognisable as an attempt at an inverse

E.g. $(\mathbf{A}^2)^{-1} = \frac{1}{\text{Their } \det(\mathbf{A}^2)}$ (A changed $\mathbf{A}^2$)

7)	$\mathbf{M} = \begin{pmatrix} 2a & 3 \\ 6 & a \end{pmatrix}$, where a is a real constant.	
(a)	Given that $a = 2$, find $\mathbf{M}^{-1}$.	(3)
(b)	Find the values of a for which $\mathbf{M}$ is singular.	(2)

Generally this was a very accessible question where the vast majority of candidates gained full marks. They had a clear understanding of the process to find the inverse matrix and were able to apply it successfully in most cases. There were some arithmetical errors in finding the determinant and some candidates could not deal with the structure of the inverse matrix. A few did not substitute the given value for a. A much longer method involving simultaneous

equations was much more prone to errors and fortunately not seen very often. In part (b) most understood the definition of a singular matrix and were able to solve their quadratic equation to give two accurate values for a . Some rejected the negative solution however, losing the last mark.

Question Number	Scheme	Marks
	(a) $M = \begin{pmatrix} 4 & 3 \\ 6 & 2 \end{pmatrix}$ Determinant: $(8-18) = -10$ $M^{-1} = \frac{1}{-10} \begin{pmatrix} 2 & -3 \\ -6 & 4 \end{pmatrix} \quad \left[= \begin{pmatrix} -0.2 & 0.3 \\ 0.6 & -0.4 \end{pmatrix} \right]$	B1 M1 A1 (3)
	(b) Setting $\Delta = 0$ and using $2a^2 \pm 18 = 0$ to obtain $a =$. $a = \pm 3$	M1 A1 cao (2) 5 marks
	Notes: (a) B1: must be -10 M1: for correct attempt at changing elements in major diagonal and changing signs in minor diagonal. Three or four of the numbers in the matrix should be correct – eg allow one slip A1: for any form of the correct answer, with correct determinant then isw. Special case: a not replaced is B0M1A0 (b) Two correct answers, $a = \pm 3$, with no working is M1A1 Just $a = 3$ is M1A0, and also one of these answers rejected is A0. Need 3 to be simplified (not $\sqrt{9}$).	